

Research assignment: module 01

The language of business analytics

Question 1

Business Analytics: refers to a way in which business use statistical methods and computing technologies for processing, mining and visualizing data to uncover patterns and insights that can help business to make better business decisions.

Business analytics clean and correct data, create data visualizations and provide insights on where the business is today while helping predict what will happen tomorrow and in future.

Example: A coffee shop analyzing sales patterns to decide which branch need more staff during Christmas festive.

Question 2

Data Analysis: is a process of collecting, transforming and organizing data in order to draw conclusions, make predictions and drive informed decision-making.

They process examine raw data to uncover patterns, draw meaningful insights and it helps individuals and businesses to understand past performance, monitor current trends and predict future outcomes.



Example: An airline company analyse past data on seat bookings, flight timing, and passenger preferences to forecast future travel trends. The predictive analysis allows the airline to optimize flight schedules and plan for peak travel periods as well as set competitive ticket prices to improve customer satisfaction and increased revenue.

Question 3

Business Intelligence: is a set of technological processes for collecting, managing and analyzing organizational data to yield insights that that inform business strategies and operations.

Business organisations can use the insights gained from business intelligence and data analysis to improve business decisions, identify problems or issues, spot market trends and find new revenue or business opportunities.

Example

Netflix checks the search history of every viewer on the platform to figure out what they like to watch. Using that information, Netflix offers suggestions of shows or movies the viewers like to watch. This save viewers from searching/researching and offer them the content of their interest, without searching for it.



This keeps people on the platform and makes them invest in their subscription plans.

Question 4

Data-driven decision making: Is an approach that emphasizes using data and analysis instead of intuition to inform business decisions.

Example: Amazon uses data to segment customers based on location and buying behaviour to build ~~targeted~~ targeted marketing campaigns.

Question 5

Big data: refers to extremely large datasets that regular tools struggle to handle, they are handled/analyzed computationally to reveal patterns, trends and associations, especially relating to human behaviour and interactions. This data comes from many places such as social media, sensors, online transactions and more.

Big data consists of three V characteristics such as Volume which is the amount of information, huge volume of data, Velocity which is the speed at which data was generated, Variety different forms in which data comes as e.g. text, images, videos and more.



Example: Retailers use big data to check/trace product demand. They analyze sales data, seasonal trends, weather forecast and social media to predict which products will sell well and when. This helps them stock shelves and avoid overstocking or running out of popular items.

Question 6

Descriptive Analytics: Uses statistical summary and data visualization techniques to describe historical data, it helps to identify patterns, trends and relationships within the data and clearly showing what happened and what is currently happening.

Example: A hospital analyzing patient records, they calculate average waiting time in the emergency room, categorize most frequent diagnosis, track monthly trends in admissions. By summarizing data and using visuals like bar charts or line graphs, they can identify patterns and understand what is happening within the hospital. This descriptive analysis will allow the hospital to focus on areas for improvement and making what is currently working more effective.



Question 7

Diagnostic Analytics:

~~Diagn~~ analytics ~~uncovers~~ focuses on uncovering the root causes, performance gaps and patterns that inform better decisions and future strategies.

Example: Healthcare industry analyzing why certain populations experience longer recovery times, identifying the factors behind high readmission rates and determining which treatment protocols yield better outcomes for specific conditions.

Question 8

Predictive Analytics: is a branch of advanced analytics that makes predictions about future outcomes by using historical data combined with statistical modeling, data mining techniques and machine learning.

Example: Clothing store do analyze past sales trends to predict seasonal demand, ensuring sufficient stock and targeted promotions

Question 9

Prescriptive Analytics: is the process of using data to determine an optional course of action. All relevant factors are considered on this type of analysis to yields recommendations for the next steps.



Example: An E-hailing business (Uber) uses prescriptive analytics to adjust ticket or uber prices in real time. As the peak time arises the system automatically increases prices and as when the uber encounter traffic on the road and the car stays or reach the destination longer than initially predicted the price of the trip increases. It also give discount to the commuters.

Question 10

Data literacy: involves a comprehensive range of capabilities, including assessing the reliability of data sources, identifying trends, patterns and outliers within datasets and clearly communicating insights to both technical and non-technical stakeholders.

Example: Company that specialize in service, they implement data-driven approach to project management, by collecting and analyzing data on project timelines, resources allocated and team performance, they are able to streamline their processes. This lead to more efficient project completion and better resource management. Data literacy will significantly enhance project outcomes.



Section 02

Data, measurement and reporting

Question 11

Structured data: is data that is organized in a predefined format, making it easily readable and understandable by both humans and machines. The structured data can be achieved through a well-defined schema or data model, where data is stored in an orderly way such as rows and columns.

Example: A customer database in a coffee shop might contain structured records with fields like name, Address, phone number, and email.

Question 12

Unstructured data: is an inconsistent format that make it difficult to store data in traditional rows and columns.

Examples: Social media posts, audio files, text document and images.

Question 13

Data warehouse: is a central repository system in which businesses store valuable information, for data analytics and ^{for} reporting purposes.

Example: Coffee shop ~~store~~ data warehouse can



Combine sales transactions, customer loyalty program details, and inventory data to understand buying habits and optimize stock management.

Question 14

Data Mining: refers to the systematic process of organizing large datasets to identify patterns, correlations or trends. Use algorithms and statistical methods to extract valuable insights from large datasets.

Example:

Q. The health sector can use algorithms to predict diseases from patients records, improving early diagnosis and treatment plans.

Question 15

Data Visualization: is a tool that translate complex datasets into visual formats, making it easier to understand, interpret, and derive actionable insights.

Example! During covid-19 pandemic data visualization was used to track and manage the spread of the virus. Governments, health organizations and research institutions used interactive dashboards to present real-time data on infection rates, recoveries and fatalities.



Question 16

Dashboard: Consolidate scattered data into a single, unified view and presenting it through interactive visuals such as charts, graphs, and tables that make sense at a glance. They allow teams to track and analyze key performance indicators (~~the~~ KPIs) and other metrics by translating complex information into a format that ~~is~~ both technical and non-technical audiences can understand.

Example: Marketing dashboard, used to measure performance across campaigns, channels, and customer touch points. Metrics include follower growth, website conversions and organic and paid ad spend.

Question 17

Metrics: Is a quantitative value that represents an amount, frequency, duration, proportion, cost or outcome. It is used to describe and compare the behavior of a website, application, campaign or process over a defined period and within a defined scope.

Examples: Monetary value recorded for purchases and subscriptions.

Distinct people who have interacted with the website or application according to the platform's criteria.



Question 18

Key Performance Indicator (KPI): is indicator/metric that is used to measure how effectively an organization, team, or individual is achieving a specific objective.

Example: Employee task performance is used to track how employees perform on each given task individually or team work. Marketing team use page views, social media followers to measure the sales drive and customer visibility and engagement.

Question 19

Benchmarks: is the process of comparing ~~the~~ organization's performance against organizations that operate in the same niche, are of similar size, and have a similar target audience. KPIs, metrics, are used as references to evaluate the performances.

Example:

Example: Human resources use benchmark metrics such as employee engagement, turnover rates, and hiring practices across different departments and organizations.

Question 20

Segmentation: is a process of breaking down dataset by to spot patterns that average the data together.



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Example: Medical aid company can segment its customers into different plans and can also be getting different rewards from being part of certain plans.

Section 03: Customer and Sales KPIs

Question 21

Conversion Rate: is a key performance indicator that measures the percentage of users who complete desired actions on a website or application.

Example: Digital marketing team can use the conversion rate to provide insight into effectiveness of marketing strategies and user experience design, a higher conversion rate indicates that a website is successfully persuading visitors to take desired actions.

Question 22

Customer Acquisition Cost (CAC): is the total cost a business incurs to acquire a new customer. This includes all expenses related to marketing, advertising and the effort of sales used to attract and convert potential customers.

Example: If a company spend R10 000 on marketing and sales that means $\frac{R10\,000}{100} = 100$

that means customer acquisition cost will be R100 per customer.



Question 23

Customer lifetime value (CLV/LTV): is the total revenue a business expects to earn from a customer during their entire relationship with the company. It helps businesses to understand the long-term value of each customer. The ratio measures the relationship between the value ~~Example~~ a customer generates and the cost required to acquire them.

Examples: Customer spends R5 000 per purchase and they buy four times a year, which is R20 000 per year. Over four years of relationship with same way of purchase they will generate R80 000 in revenue from this customer. When we subtract R20 000 in cost to serve of the period and the customer lifetime value will be ~~R80 000~~ R60 000.

Question 24

Churn rate: is the percentage of customers or subscribers that stop their relationship with a business over a specific period, indicating the company's ability to retain clients and overall health.

Example: An insurance company acquire 1000 new policies in one ^{year} month. That same ^{year} month the company lose 120 policies.

The churn rate would be:

$$\begin{aligned} 120 / 1000 &= 0.12 \times 100 \\ &= 12\% \end{aligned}$$



Question 25

Customer Retention: is the rate at which customers continue to purchase goods or services from a business over time. Analysis of key metrics helps to identify trends and refine retention.

Example: A Customer recently purchased new music equipment. A few days later, a Customer Service rep reached out to check if everything was working well and offered a quick guide on how to use the equipment's advanced features.

This personal touch made the Customer feel valued and ensured they were getting the most out of their music equipment, making them more open to upselling.

Question 26

Net Promoter Score (NPS): is a customer loyalty and satisfaction measurement taken by asking customers how likely they are to recommend products or services to others. NPS can be used to predict business growth.

Example: Telecommunication like MTN create survey after each client service given from the sales agent, this help the company to see how the service is to their clients and how satisfied is for it. If the service given to their clients is of satisfactory.



Question 27

Average Order Value: Is the average amount of money a Customer spends per order. It helps with understanding purchasing behavior, optimize pricing and design strategies to increase revenue per order.

Example: Company in retail industry ~~use~~ ^{make} a revenue of ₹5000 per day with 100 transactions, the average order value is ₹50 per transaction.

Question 28

Sales funnel - is a visual representation of the stages in which a potential customer goes through, guiding them towards becoming a paying customer. The funnel will help the business to effectively manage and optimize their sales cycle.

Example: An online business like takealot use online visibility, search engine optimization to optimize the content and rank higher search engine results to lower results. they use the results to capture customers by showing things that can interest them to buy. they offer multiple payments options such as credit cards, pay just now. They sent regular marketing advertisement to keep ~~for~~ ^{them} into the business.



Section 04: Marketing and Campaign

Question 29

Marketing Campaign Analytics! is a strategic effort that businesses or organizations take to promote products, services or message to a targeted audience.

Example: Coke share campaign, the campaign replaced / ~~the deleted~~ the people's names on their bottles, popular names was used, the campaign was encouraging people to share Coke with their friends and loved ones.

Question 30

Post-campaign Analysis! is the analysis that help organisation into turning raw data to actionable insights, it improve resource allocations and help organisation to ~~analyse~~ ^{focus on} what works best, spend money in a smarter way and learn from past successes and failures.

Example! After coke's share campaign, it will compare the rate at which customers buy their product before and after the campaign. If lets say the selling rate of coke has increased by 20% post campaign, this will be deemed as a successful campaign.



Question 31

Incremental Revenue: is the money that is generated from an additional sales quantity, it is used to analyze and compare the revenue generated by two different strategies.

Example: A coffee shop can offer few dozen of syrup and milk options, they encourage customers to spend more money per drink. The coffee can move from being R3 to R5 with add-ons, this create increment to the revenue.

Question 32

Return On Investment: is a metric that is used to measure the return or gain that is generated from an investment in relation to its cost in a business.

Example: A company that sell coffee invest R10 000 in a marketing campaign which resulted in generating R100 000 in additional sales revenue, this will create ROI of 100% indicating that the marketing campaign generated a profit equal to the initial investment.



Question 33

Return on Ad Spend (ROAS) is a metric that is used for advertising performance, to demonstrate how much revenue is earned from ad spend. It helps to compare Campaigns and Channels.

Example: If a Youtube Platform Spend R5000 on ads campaign and generates R20000 in sales that was directly attributed to those ads, this will give a ratio of 4:1 Roas.

Question 34

Click-through rate (CTR) is a metric that is used to measure the percentage of users who click the specific link out of the viewers who are able to see the ad or webpage.

Example: Youtube use advertisements (ads) in the middle of videos posted, lot of people who ^{see} those videos will either click on the ads while some don't click at all.



Question 35

Bounce rate is a metric that measures the percentage of the people who cancel or move out of the webpage after only viewing a single page.

Example: Company can be able to see that after people click their webpage there is not enough information or the images are not showing well that give customer less to no interest to further purchase on their webpage.

Question 36

A/B testing is a metric that is used by marketers to split their audience and create multiple versions of a particular webpage / product in a campaign to test how effective is the page / product and determine which version performs better.

Example: A concert industry that sell a company that is hosting a concert is selling tickets and two landing page for sales is created for early birds and normal tickets this A/B testing is used to test the sales page for tickets.



Question 37

Lead Generation: is a component of marketing that is used to focus on attracting and capturing the potential customers that will be able to drive business growth. They capture customer's interest in a product or services, the goal is to make use of this interest to make sales.

Example: Company like ~~Africa~~^{Flysaair} Flysaair create a flight ticket sale of R9 per ticket for trips such as Durban and more. This makes customers who enjoy holidays to be able to join the sale of tickets and this drive a high volume of sales to the company.

Question 38

Marketing attribution: is a process of identifying the marketing touchpoints which has created or contributed conversion and give each credit, this help to identify the channels that bring highest quality customers, it helps organisation to stop overspending on channels that are not producing customers/sales.



Example: A Customer discover a brand through an 'influencer's' vlog post on tiktok ad and click ^{link} which convert through a google search ad and give last click to google search ad.



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